

TAPOR: 3D Hand Pose Reconstruction with Fully Passive Thermal Sensing for Around-Device Interactions

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This paper presents the design and implementation of TAPOR, a privacy-preserving, non-contact, and fully passive sensing system for accurate and robust 3D hand pose reconstruction for around-device interaction using a single low-cost thermal array sensor. Thermal sensing using inexpensive and miniature thermal arrays emerges with an excellent utility-privacy balance, offering an imaging resolution significantly lower than cameras but far superior to RF signals like radar or WiFi. The design of TAPOR, however, is challenging, mainly because the captured temperature maps are low-resolution and textureless. To overcome the challenges, we investigate thermo-depth and thermo-pose properties, proposing a novel physics-inspired neural network that learns effective 3D spatial representations of potential hand poses. We then formulate the 3D pose reconstruction problem as a distinct retrieval task, enabling accurate hand pose determination from the input temperature map. To deploy TAPOR on IoT devices, we introduce an effective heterogeneous knowledge distillation method, reducing computation by 377×. TAPOR is fully implemented and tested in real-world scenarios, showing remarkable performance, supported by four gesture control and finger tracking case studies. We envision TAPOR to be a ubiquitous interface for around-device control and have open-sourced it at <https://github.com/aiot-lab/TAPOR>.

CCS Concepts: • **Computer systems organization** → **Embedded and cyber-physical systems**; • **Networks** → Hardware Sensor applications and deployments;

Additional Key Words and Phrases: 3D hand pose, Gesture recognition, Thermal array, Thermal sensing

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1 Introduction

As the Internet of Things (IoT) devices and edge computing continue to proliferate globally, the need for seamless and efficient interaction has been growing rapidly. Around-device gesture interaction [37, 44] has become an integral part of modern interactive systems, providing an increasingly essential interface for applications such as smart homes, autonomous vehicles, virtual and augmented reality environments, human-robot interaction, and wearables/hearables, etc.. Around-device gesture interaction, focusing on close-range sensing, enables users to control many devices in the same environment using a consistent set of touchless gestures. This capability is

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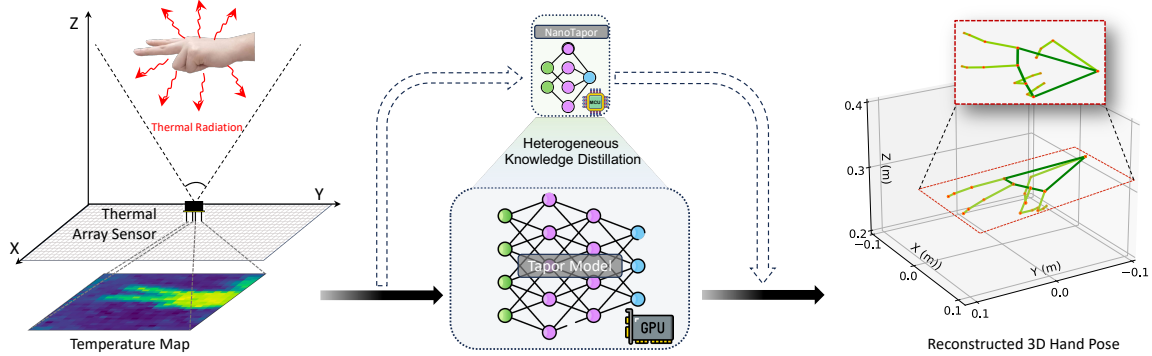


Fig. 1. Overview of TAPOR. TAPOR reconstructs the 3D hand pose from a temperature map generated by a single low-cost thermal array sensor through a physics-inspired neural network model. Leveraging the distilled NanoTapor model, the system operates in real-time on IoT devices with negligible performance degradation, making it suitable for a wide range of around-device interaction applications.

particularly crucial for resource-constrained IoT devices, which often lack traditional input mechanisms such as touchscreens or keyboards, thereby driving the rapid expansion of the gesture control market, which is projected to exceed \$57 billion by 2033 [25].

Despite its potential, enabling ubiquitous around-device gesture interaction is fraught with challenges, including constraints related to privacy, accuracy, computational complexity, cost, and power consumption. Significant efforts have been undertaken to address these challenges, with notable contributions from both industry and academia. Commercial solutions such as Google Soli [37] and Vtouch [74], alongside research initiatives [47, 69, 77, 81], have made strides in advancing gesture recognition technologies. However, many existing approaches are limited to recognizing a predefined set of gestures, which inherently restricts their flexibility in supporting user-defined gestures. Recently, 3D hand pose reconstruction using various technologies has gained significant attention as a key enabler for more flexible and scalable gesture control. Despite notable advances, existing solutions face several limitations in the context of ubiquitous around-device gesture interaction. For example, wearable systems create adherence issues as users are required to wear specific devices, in addition to an implicit gap in integration into various IoT [7, 84]. Camera-based approaches raise significant privacy concerns and are usually computation-intensive [1, 56, 87]. Wireless sensing solutions using Wi-Fi [19, 39] or millimeter-wave [35, 37] signals preserve privacy, yet suffer from deficient domain generalization [85], and cumbersome hardware setup (e.g., a pair of separately deployed Wi-Fi transceivers) [27]. Similarly, acoustic methods face comparable issues and are additionally vulnerable to interference from environmental noise [30, 44].

In this paper, we present the design, implementation, and deployment of TAPOR, a **Thermal Array-based 3D hand Pose Reconstruction** system that is privacy-preserving, fully passive, accurate, and cost-effective. TAPOR explores and exploits thermal sensing for 3D hand pose reconstruction using a low-cost thermal array. As shown in Fig. 1, a thermal array is a miniature sensor that captures thermal radiation from a user's hand. Compared with the aforementioned modalities, a thermal array produces frames with magnitude lower imaging resolution (e.g., 32×24 thermal pixels) than RGB/depth/thermal cameras, while offering significantly higher resolution than RF or acoustic signals, striking an excellent balance between utility and privacy. As further detailed in Tab. 1, since the sensing signal, i.e., thermal radiation, is emitted directly by the target (i.e., the hand), TAPOR is a *fully passive sensing* system that requires no external signal sources. It can operate in darkness, unlike vision-based methods, and consumes less power compared to mmWave radars and WiFi devices. Built upon thermal array-based sensing, TAPOR achieves accurate and robust 3D hand pose reconstruction, offering a scalable and flexible solution for

Table 1. Contactless hand pose reconstruction techniques.

Sensing Modality	Privacy Protection	Sensing Mode	Sample Device	Power	Cost
RF Devices	Strong	Active	IWR1843 Radar	>2 W	~299\$
Acoustic Devices	Weak	Active	miniDSP UMA-8-SP	>20 W	~155\$
RGB/IR Cameras	Very Weak	Quasi-passive	Basler IMX334	>2.7 W	~199\$
Thermal Array	Medium	Fully Passive	Melexis MLX90640	<0.08 W	~20\$

hand gesture control. Notably, TAPOR requires only a single tiny thermal array, making it a low-cost and efficient solution that can be easily integrated into various IoT devices.

However, 3D hand pose reconstruction from a temperature map produced by the thermal sensors using a lightweight model poses significant challenges. First, thermal arrays are fully passive sensors, capturing only the thermal radiation emitted by the target without transmitting any signals. Therefore, depth/position estimation from the captured temperature readings is exceedingly challenging due to the lack of a reliable model correlating thermal radiation with distance. The difficulty is exacerbated by the use of a single, low-resolution thermal sensor. Hence, the problem necessitates a novel solution different from existing methods designed for depth or RGB cameras, which either use dedicated infrared lights for ranging [82] or rely on high-resolution textured features [80]. Second, key point detection from the low-resolution, textureless temperature maps is far from straightforward. Despite extensive research on low-resolution images or textureless data (e.g., depth cameras or lidar point clouds) separately, low-resolution coupling with the lack of texture complicates key point identification for various hand poses. Last, it is non-trivial to build an efficient end-to-end system for real-time deployment on IoT devices with highly constrained resources (e.g., ESP32 with an MCU).

TAPOR presents two unique designs to overcome these challenges and deliver a practical 3D hand pose reconstruction system deployable on IoT. First, we propose a novel physics-inspired neural network that formulates the 3D hand pose reconstruction as a retrieval problem, enabling accurate and robust 3D hand pose estimation from a single temperature map. Based on an in-depth understanding of thermal data from the principles of thermography, we identify a distinct opportunity to learn from the thermo-depth relationship and the 2D thermo-pose context in conjunction with hand structure knowledge. Our design incorporates (i) a thermo-space encoder that integrates the thermo-depth property and 2D thermal imaging context to construct a 3D spatial representation for hand poses; (ii) a key point (KP) feature extractor that generates 2D KP features by incorporating common knowledge of hand structure and temporal coherence; and (iii) a position retrieval decoder that retrieves the 3D hand pose by matching the KP features (constraints) to the 3D spatial representation (solution space).

Second, we propose a heterogeneous knowledge distillation approach to transfer the physics-inspired model into an extremely lightweight network for IoT deployment. Specifically, rather than involving attention blocks, we constrain the model structure to purely convolutional neural networks (CNNs), which are efficiently supported by edge devices. We also perform feature dimension reduction to minimize computational overhead. To transfer knowledge between models of different structures and feature dimensions, we propose heterogeneous knowledge distillation with a learnable adapter to align structures and feature dimensions, significantly reducing the computation and successfully delivering a tiny version of TAPOR, named NanoTapor, for real-time on-device deployment for IoT.

We implement TAPOR as an end-to-end prototype using a Melexis MLX90640BAB thermal array (32×24 pixels, \$20) and conduct experiments in diverse real-world settings. We collect a dataset from 10 users across 5 environments, yielding over 60,000 hand pose samples under varied lighting, hand coverings, distances, and ambient temperatures. Our results show that TAPOR achieves a mean joint position error of 2.26 cm, sufficient

for many around-device gesture interaction applications. Notably, NanoTapor reduces computational overhead by $377\times$ compared to the original Tapor model, with only a slight increase in mean error to 3.26 cm, supporting real-time IoT operations. To further highlight TAPOR's IoT applicability, we deploy it on an ESP32-S3 device and conduct four case studies: wearable AI pin control, online meeting gestures, bathroom appliance control, and in-car gestures, all demonstrating promising performance. We foresee TAPOR as a ubiquitous interface for an interactive Internet of Thermal Things.

In summary, this paper contributes the following: (1) TAPOR, a privacy-preserving, non-contact, passive system for robust 3D hand pose reconstruction in IoT using a single, low-cost thermal sensor; (2) a physics-inspired neural network utilizing thermo-depth relationship and 2D thermal context with hand structure knowledge for 3D pose reconstruction from temperature maps; (3) a heterogeneous knowledge distillation approach, reducing computational load by $377\times$ to enable NanoTapor for real-time IoT deployment; (4) a complete implementation and extensive experiments with real-world case studies highlighting exciting applications.

The rest of this paper first presents a primer on thermal array sensing (§2), then introduces TAPOR's design (§3), followed by system implementation (§4) and experimental evaluation and case studies (§5). We discuss future work in §6, review related work in §7, and conclude the paper in §8.

2 Primer on Thermal Array Sensing

We first present a primer on thermal array sensors, with a preliminary experiment (§2.2) to understand the distinct properties of temperature maps and an in-depth understanding of thermography (§2.3) to gain insights for TAPOR design.

2.1 Thermal Array

A thermal array is a small sensor comprising a miniature array of thermal detectors arranged in a grid pattern. Each detector captures the thermal radiant power Φ emitted by objects within its field of view (FOV). The sensor converts the captured thermal radiation into an estimated surface temperature for each target area, resulting in a temperature map as illustrated in Fig. 2a. Designed in a compact size (approximately 12 mm in height and 9.5×9.5 mm in footprint), thermal arrays usually have a resolution at least two orders lower than mainstream thermal cameras and are correspondingly significantly cheaper. For example, TAPOR utilizes the Melexis MLX90640BAB thermal array sensor, which features a resolution of 32×24 and costs about US\$20, with a reported temperature accuracy of $\pm 1.5^\circ\text{C}$, a FOV of $35^\circ \times 55^\circ$, and a frame rate up to 64 Hz. Thanks to the low price and compact size, thermal array sensors emerge as an attractive sensing modality, especially for IoT devices.

2.2 Investigative Experiment

Computer vision techniques for hand pose reconstruction from RGB images have reached a high level of maturity in recent years. Consequently, a natural approach is to leverage existing RGB-based models, such as Mediapipe [83], and apply them directly to temperature maps. However, this strategy proves to be suboptimal, as temperature maps exhibit distinct characteristics that differ significantly from RGB images. These unique properties, which are explored in detail in the following investigation, hinder the direct applicability of existing RGB-based models and necessitate specialized solutions tailored to the thermal domain.

Our preliminary study involves five participants, four males and one female, who performed various hand poses in front of a thermal array sensor and an RGB camera to collect thermal and RGB sample pairs, as shown in Fig. 2a. In total, over 1000 samples were collected from each participant. To examine the effectiveness of existing RGB-based models, we employ Mediapipe [83], a well-known and widely used model for 2D hand pose estimation. The hand pose is represented by 21 key points, as depicted in Fig. 3a. For a comprehensive evaluation, we retrain and test the MediaPipe model under three input conditions: 1) the original RGB images (256×256), 2)

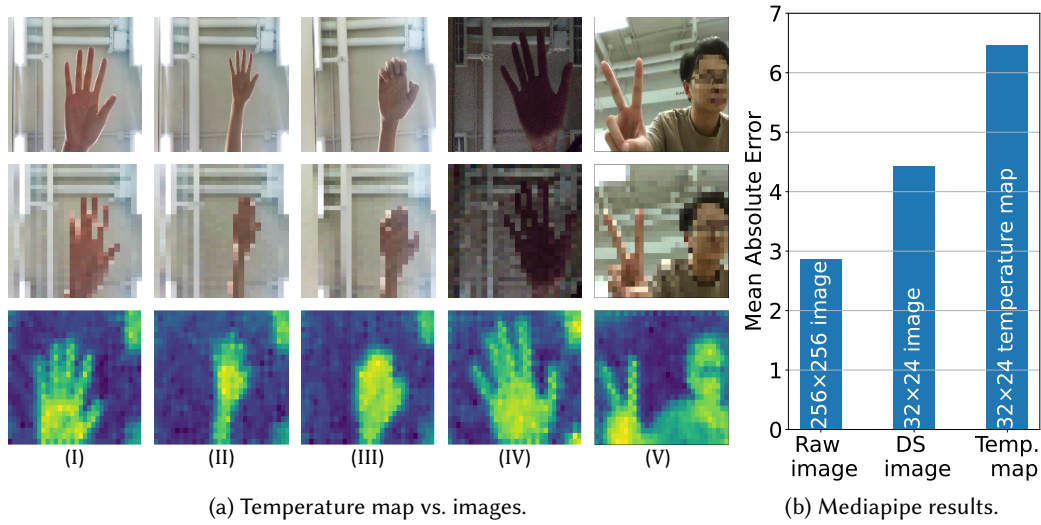


Fig. 2. Modality comparison. (a) Example samples highlight distinct properties of the temperature maps: (i) single-channel, (ii) low-resolution, (iii) textureless, (iv) illumination immunity, and (v) privacy-preserving. (b) Performance comparison of Mediapipe on original RGB images (raw image), downsampled (DS) images, and temperature maps.

the downsampled RGB images (32×24), and 3) the thermal array temperature maps (32×24). All models were trained and evaluated using the same 7:1:2 train/val/test dataset split. As illustrated in Fig. 2b, the results show that, Mediapipe performs considerably worse on the thermal array data than on RGB images, with a performance drop of up to 100%. This suggests that existing RGB-based pose estimation models have difficulty adapting to thermal array data, highlighting the need for a novel model design.

Based on an in-depth investigation of the temperature maps reported by the thermal array sensor, we identify several distinct properties that lead to the unsatisfactory performance of existing RGB-based models, which are designed for very different RGB images. As illustrated in Fig. 2a, the temperature maps exhibit the following unique characteristics: 1) **Low resolution**: Thermal arrays typically generate a temperature map of 32×24 (or even fewer) pixels. While the low resolution inherently preserves privacy, most prior CV models assume a minimum resolution of 128×128 pixels [11, 14, 38]. 2) **Textureless**: As opposed to visual images, the temperature maps lack texture information, making it more difficult to identify overlapped objects (e.g., crossed fingers as in Fig. 2a(iii)). 3) **Single channel**: The temperature map possesses only one channel characterizing the captured thermal radiation from the target, unlike the multi-channel RGB images. The combination of these properties makes temperature maps a unique data type, posing significant challenges for 3D hand pose reconstruction using thermal arrays, a novel problem that requires a new solution.

Despite these challenges, thermal arrays offer many advantageous features for practical hand pose reconstruction, as they are **low-cost**, **privacy-preserving**, and, importantly, **illumination-immune**. This motivates us to overcome these challenges and design TAPOR for 3D hand pose reconstruction using a thermal array sensor.

2.3 Thermo-Pose Modeling

We investigate the principles of thermography theory [42] to see if it is possible to derive depth and position directly from the temperature readings, an essential component for 3D hand pose reconstruction.

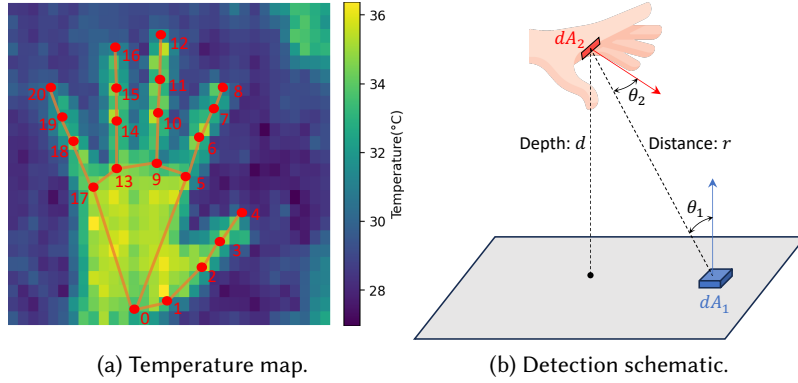


Fig. 3. Temperature map and thermal radiation detection diagram. (a) Temperature map of an outstretched hand, with red dots marking key points. (b) The thermal radiation detection schematic of the thermal detector.

Thermo-Depth Relation: Thermography theory explains how a thermal detector measures the radiant power emitted from a target, which is then converted to temperature readings. As shown in Fig. 3b, the measured radiant power $d\Phi$ of the thermal radiance from a unit surface area dA_2 to a unit detector area dA_1 is expressed as [5]

$$\Phi(d) = \epsilon e^{-\gamma d / \cos \theta_1} L \cdot A_1 \int_{A_2} \frac{\cos \theta_1 \cos \theta_2}{(r / \cos \theta_1)^2} dA_2, \quad (1)$$

where L is the radiance from the source, r is the distance between surfaces dA_2 and dA_1 , θ_1 and θ_2 are angles between the normal vectors of the surfaces and the radiation direction. γ is the attenuation coefficient, and ϵ denotes the emissivity of the target, *i.e.*, human hand.

Ideally, Eq. (1) should provide an opportunity to derive depth from the measured temperature T , given that T can be calculated from and is proportional to the captured radiant power, *i.e.*, $T \propto \sqrt{\Phi}$. In practice, however, this intuitive approach does not work. The main reason is that the source radiance L and the effective radiant area A_2 heavily depend on the target temperature, surface coverings, environmental heat interference, etc, all time-varying and unpredictable. Additionally, measurement noise in low-cost thermal sensors adds further complexity. Specifically, as shown in Fig. 4a, we make the following observation:

Observation I: The captured thermal radiant power has a non-linear and negative correlation with hand depth.

This consistent, monotonous thermo-depth relationship, when combined with the imaging resolution, presents an opportunity to address the problem via data-driven learning (§3.2.1).

Thermo-Pose Relation: The above analysis models the hand as a single body. To further understand the relationship, we dive into the detailed hand pose structure and model its impact on the captured temperature map. Recalling Eq. (1) and the relation of $T \propto \sqrt{\Phi}$, the temperature readings for different key points of the hand should be similar, since they are all at approximately the same depth. As shown in Fig. 4a, however, both the temperature readings and the changing trends with respect to depth vary significantly across different key points.

To understand the root causes, we can formulate the impacts of depth on different key points as follows. As shown in Fig. 4b, for key points at the edge of the hand, depth changes do not significantly affect the total radiant surface area A_2 . Thus, according to Eq. (1), the captured temperature decreases with respect to d as follows:

$$\frac{\hat{T}(d_1)}{\hat{T}(d_0)} \propto e^{-\gamma / \cos \theta_1 \frac{d_1}{d_0}} \cdot \frac{d_0^2}{d_1^2}, \quad (2)$$

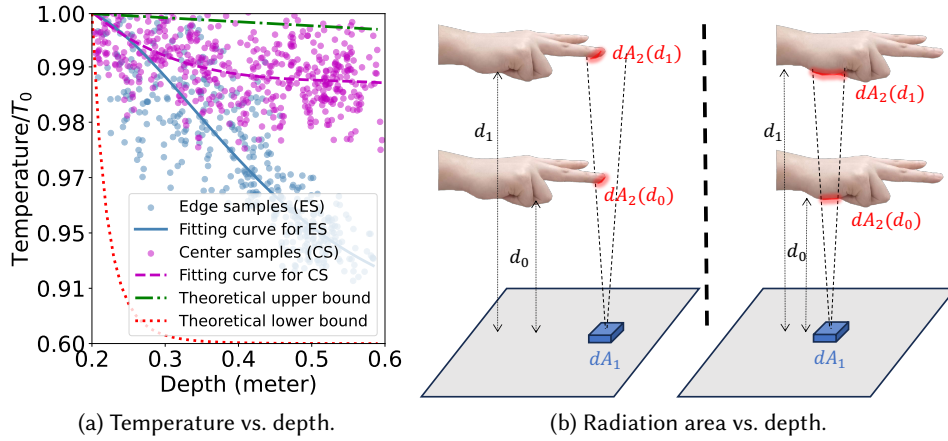


Fig. 4. Thermo-depth analysis. (a) The detected temperature of hand key points overall decreases, at a large scale, regarding depths. Yet the decreasing trends vary significantly, depending on the locations of the key points relative to the hand. T_0 is the detected temperature at a reference depth of 0.2 meters. This phenomenon is explained in (b): The radiation area A_2 increases with larger depth for center key points, compensating for the increased air attenuation. The radiation area, however, remains nearly constant for edge key points, leading to more drastic temperature decreases.

Conversely, since the radiant surface area A_2 increases proportionally to d^2 , the captured temperature of key points in the center of the hand is less impacted by depth and changes differently as:

$$\frac{\hat{T}(d_1)}{\hat{T}(d_0)} \propto e^{-\gamma/\cos\theta_1 \frac{d_1}{d_0}}. \quad (3)$$

Fig. 4a demonstrates both the theoretical bounds and practical measurements of these highly diverse trends, where the theoretical bounds are obtained based on [42] by setting $d_0 = 0.2m$, $\theta_1 = 0$, and $\gamma = 0.025$. On this basis, we make another key observation as follows:

Observation II: *The captured temperature maps exhibit distinct 2D spatial patterns for different hand poses with different key point structures, even at approximately the same depth.*

Therefore, subtle variations corresponding to the hand region in the captured temperature maps will imply changes in hand poses, allowing rich latent information for learning the detailed pose structure (§3.2.2).

In the next section, we explore leveraging these observations in a novel neural network to learn positional information and key point features for 3D hand pose reconstruction.

3 Physics-inspired Design

This section presents our physics-inspired neural network design, followed by its ultra-lightweight variant for mobile and IoT devices.

3.1 Overview

Tapor model consists of three main components: Thermo-Space Encoder, Key Point (KP) Feature Extractor, and Position Retrieval Decoder, as depicted in Fig. 5.

Like RF ranging, thermal radiation attenuation, as described in Observation I in §2.3, can be utilized for range/depth estimation. However, direct depth estimation based solely on temperature values results in large errors, e.g., about $\pm 20cm$ for human bodies [45, 86], which can be even worse for the significantly smaller hands.

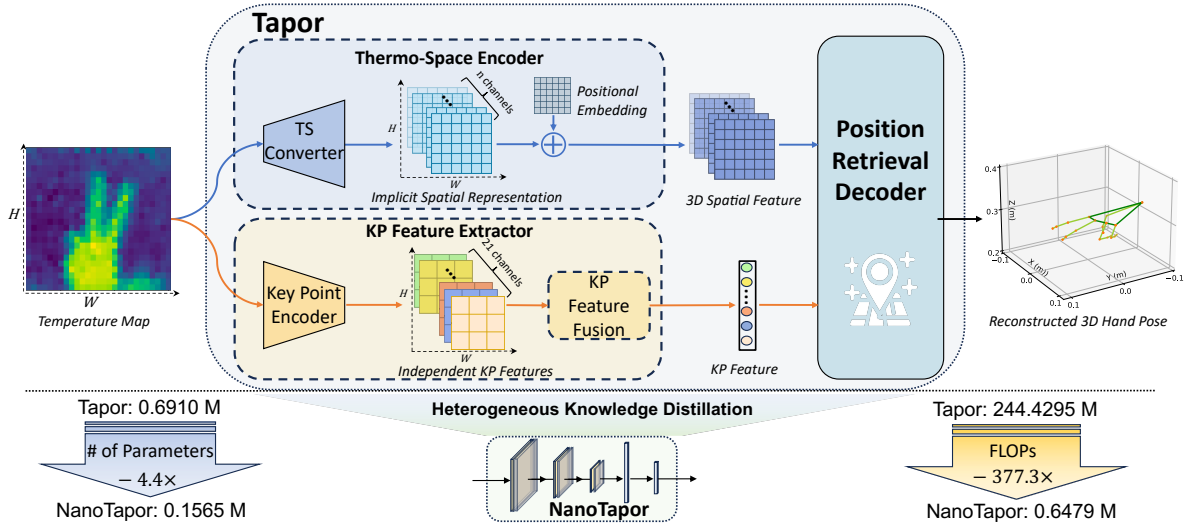


Fig. 5. A structure overview of the Tapor and NanoTapor models.

In TAPOR, we exploit a complementary opportunity thanks to the 2D imaging capability of the thermal arrays. By leveraging the contextual information (e.g., 2D hand shape) in the low-resolution temperature map, we establish a context-aware mapping, called the thermo-space encoder, that converts detected hand temperature into a 3D spatial representation of the hand's depth and all possible poses. This approach reduces the depth estimation error to around 2cm, as demonstrated in our experiments (§5).

On this basis, unlike previous works that mostly consider a regression [33, 48, 83] or detection problem [54, 67], we formulate the 3D hand pose reconstruction as a *retrieval task*. To reconstruct the 3D hand pose, we first draw inspiration from Observation II (§2.3) and build a Key Point Feature Extractor to extract key point features from the 2D temperature maps with the help of common hand pose knowledge. We then design a Position Retrieval Decoder to utilize the key point features as queries to retrieve the 3D position of each key point from the 3D spatial representation. At a high level, this process can be viewed as finding the unique solution (i.e., the output 3D hand pose) from the possible solution space (i.e., the 3D spatial representation) with constraints (i.e., the key point features). By doing so, TAPOR reconstructs the 3D hand pose $J \in \mathbb{R}^{21 \times 3}$, represented by the 3D physical positions of 21 key points, from the temperature map $T \in \mathbb{R}^{32 \times 24}$.

3.2 Tapor Model Design

3.2.1 Thermo-Space Encoder. The thermo-space encoder is dedicated to constructing a representation of all possible 3D hand poses from the current temperature map. To achieve this, we introduce a Temperature Space (TS) converter that transforms the temperature values and contextual information into an implicit spatial representation of possible hand locations and poses. By integrating a positional embedding with the implicit spatial representation, the thermo-space encoder produces a 3D spatial feature that describes the potential solution space of the hand pose.

The TS converter, represented by $E_{ts}(\cdot)$, extracts information from the input temperature map, T . For model efficiency, we employ the inverted residual block from MobileNetV2 [60] to construct the converter. However, the convolution operation reduces the H and W dimensions of T as shown in Fig. 5. To counter this, we use an upsample layer to scale up the original dimensions along the H and W axes, with an upscaling factor α . The

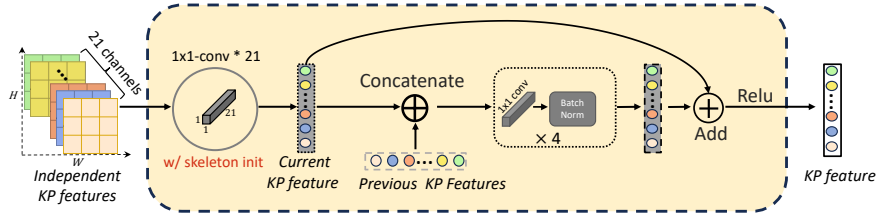


Fig. 6. The KP feature fusion module. This module facilitates information exchange across different key points and includes an optional residual connection for feature fusion from previous inputs.

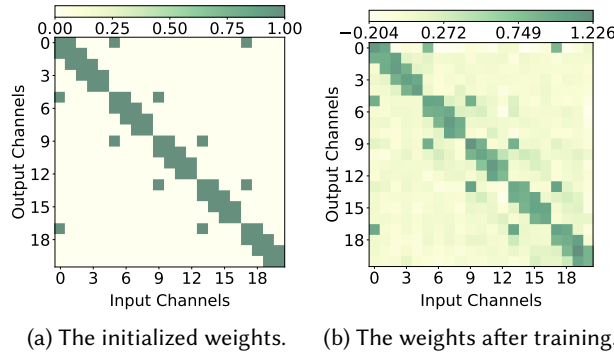


Fig. 7. Skeleton init. (a) Initial weights of the 1x1 convolutional kernel based on hand structure. (b) The learned weights after training retain hand structure knowledge.

output of the TS converter is the implicit spatial representation, denoted as $f_{ISR} \in \mathbf{R}^{n \times 15 \times 20}$, where $n = 128$ is the number of channels. The 3D spatial feature is then produced as $f_{spatial} = E_{ts}(T) \oplus P_e$, where P_e is the positional embedding [73], $\oplus$ represents element-wise addition, and $f_{spatial} \in \mathbf{R}^{128 \times 15 \times 20}$. Thus, the input 32×24 2D temperature space is extended to the 3D spatial feature space with dimensions $128 \times 15 \times 20$.

3.2.2 KP Feature Extractor. Extracting distinct features for all key points is essential for accurately retrieving their locations from the 3D spatial feature space. We first introduce a key point encoder to derive the feature of each key point directly from the entire temperature map, capturing both hand shape and temperature information. To address the challenges of occluded key points and reconstructing a realistic hand pose, we design an efficient KP feature fusion module. This module compensates for missing information in occluded key points by leveraging features from other key points and incorporating information from previous inputs. It further enhances the coherence of the hand structure by incorporating prior knowledge of hand anatomy, enabling the extraction of all key point features without the need to see all fingers.

Specifically, the key point encoder, $E_{kp}(\cdot)$, shares a similar structure with the TS converter but with two key differences: (1) it does not need to preserve the H and W dimensions, so it skips the upsample layer, and (2) each channel in its output feature maps represents the features of a corresponding key point, with the total number of channels equal to the number of key points, which is 21. The encoder produces independent key point features as $f_{ikp} = E_{kp}(T)$, where $f_{ikp} \in \mathbf{R}^{21 \times 6 \times 8}$. In the KP feature fusion module, $E_{kpf}(\cdot)$, as depicted in Fig. 6, the independent key point features f_{ikp} are first processed with a 1×1 convolution κ to enable information

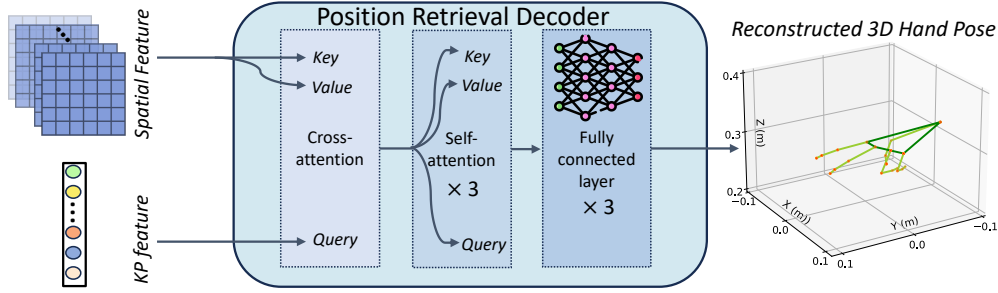


Fig. 8. Position retrieval decoder structure.

exchange. Additionally, four cascaded convolutional blocks merge the current feature with preceding features, incorporating temporal coherence. Given the sensor's variable refresh rate, a residual connection is introduced to make this component optional without altering the model's overall structure. Eventually, the KP feature extractor outputs the distinct features, $f_{kp} \in \mathbb{R}^{21 \times 48}$, of the 21 key points.

To inject hand structure knowledge, we propose a *skeleton init* method to initialize the convolutional kernel κ . As shown in Fig. 7, we tailor the kernel weights to the hand's structure by assigning a value of 1 to diagonal elements and elements corresponding to adjacent key points, while other elements are set to 0. After training, the learned weights maintain this pattern, effectively preserving the hand's structural information. The skeleton init technique boosts Tapor's performance without additional computational costs, outperforming random initialization, as evidenced in §5.4.

3.2.3 Position Retrieval Decoder. The position retrieval decoder aims to accurately identify the position of each key point. As shown in Fig. 8, it features a cross-attention layer, $D_{cross}(\cdot, \cdot)$, which uses the KP feature f_{kp} as a query to identify its corresponding location feature f_{loc} within the 3D spatial feature $f_{spatial}$, i.e., $f_{loc} = D_{cross}(f_{spatial}, f_{kp})$. Then, three self-attention layers, $D_{self}(\cdot)$, fuse the location features of all key points to produce the pose feature, $f_{pose} = D_{self}(f_{loc})$. Finally, three cascaded fully connected layers, $L(\cdot)$, convert the pose feature into 3D physical positions of all hand key points, reconstructing the hand pose $\hat{J} \in \mathbb{R}^{21 \times 3}$.

To understand whether the model functions as intended, we inspect the model in the following two aspects: (1) We assess the model's operation to ascertain whether it concentrates on the hand region (i.e., the context). As illustrated in Fig. 9a, the attention map of the cross-attention layer clearly demonstrates the model's focus on the hand region, signifying that the learned model is context-aware. (2) Moreover, to confirm whether the key point features possess distinct semantic meanings (i.e., each feature corresponding to a specific hand key point), we showcase attention maps for various hand parts generated using the relevant key point features, as depicted in Fig. 9b. The attention maps distinctly highlight their target hand regions. Additionally, the attentions to different fingers and the root point consistently emphasize the palm's center, which aligns with our objective to reconstruct an accurate and coherent 3D hand pose.

Training Loss: To train Tapor, we use two loss functions: the mean square error (MSE) loss and the bone length (BL) loss [64]. The MSE loss, $\mathcal{L}_{mse} = \frac{1}{N} \sum_{i=0}^N \|J - \hat{J}\|_2^2$, where N is the number of training samples, will cause the reconstructed pose to collapse around the palm's center. To address this, we add the BL loss, $\mathcal{L}_{bl} = \frac{1}{N} \sum_{i=0}^N \|\mathbf{b}_i - \hat{\mathbf{b}}_i\|_2^2$, where $\mathbf{b}_i$ and $\hat{\mathbf{b}}_i$ are the bone lengths of the ground truth and reconstructed poses, respectively. Thus, the overall loss function is defined as follows for end-to-end training of the model:

$$\mathcal{L}_{Tapor} = \mathcal{L}_{mse}(\hat{J}, J) + \mathcal{L}_{bl}(\hat{J}, J). \quad (4)$$

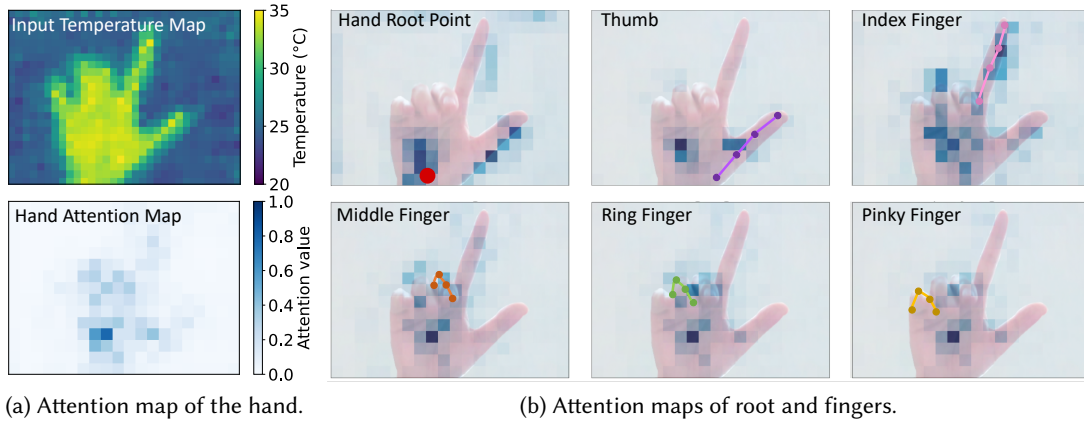


Fig. 9. Attention maps of TAPOR. (a) The attention map of the cross-attention layer highlights the model’s correct focus on the hand region, indicating effective context awareness. (b) The attention maps demonstrate that the key point encoder effectively interprets the semantic meanings of the input temperature map. Note that the RGB images are included solely for visualization purposes.

3.3 NanoTapor: Extending to the IoT

Despite Tapor’s lightweight design with fewer parameters and floating-point operations per second (FLOPs) than all baseline models, as shown in §5, it remains too large for deployment on IoT devices like the ESP32 MCU. To enable TAPOR on IoT devices and ensure real-time operation, we introduce a novel method of *heterogeneous knowledge distillation* to deliver a tiny version named NanoTapor.

Inspired by EdgeSAM [89], NanoTapor is designed as a purely CNN-based architecture, making it well-suited for edge devices. Specifically, NanoTapor comprises four convolutional layers as the encoder $E_{nano}(\cdot)$ to extract features f_n from the input temperature map T , followed by a fully-connected layer $D_{nano}(\cdot)$ to transform the features into the final pose, *i.e.*, $\hat{J} = D_{nano}(f_n)$.

To transfer knowledge from Tapor to NanoTapor, we employ knowledge distillation [23] by adding an extra feature distillation loss to the Tapor loss $\mathcal{L}_{Tapor}$:

$$\mathcal{L} = \mathcal{L}_{mse}(\hat{J}, J) + \mathcal{L}_{bl}(\hat{J}, J) + \mathcal{L}_{mse}(f_n, f_{Tapor}), \quad (5)$$

where f_{Tapor} is the feature from the Tapor model used as supervision. However, standard knowledge distillation requires matching dimensions for f_n and f_{Tapor} , which restricts the room for reducing parameters and computational overhead for NanoTapor and further hinders the ability to operate in real-time on IoT devices. For instance, processing one input temperature map takes more than 1.5 seconds on the ESP32S3 MCU, which is impractical for most interactive applications.

To tackle this issue, a straightforward solution is to reduce the dimension of f_n . This necessitates addressing the challenge of transferring knowledge from the teacher model, Tapor, to the student model, NanoTapor, despite their heterogeneous feature dimensions and structures. To overcome this challenge, we introduce an adaptor $A : \mathbb{R}^{21 \times 48} \rightarrow \mathbb{R}^{21 \times 16}$ to align the Tapor and NanoTapor models. We explore two types of adaptors: a non-learnable dimension reduction algorithm, namely principal component analysis (PCA), and a learnable linear model. Due to NanoTapor’s heterogeneous structure, the PCA method fails to adapt and leads to performance degradation, as demonstrated in §5.4. Consequently, we opt for a learnable linear layer as the adaptor $A(\cdot)$, which offers superior

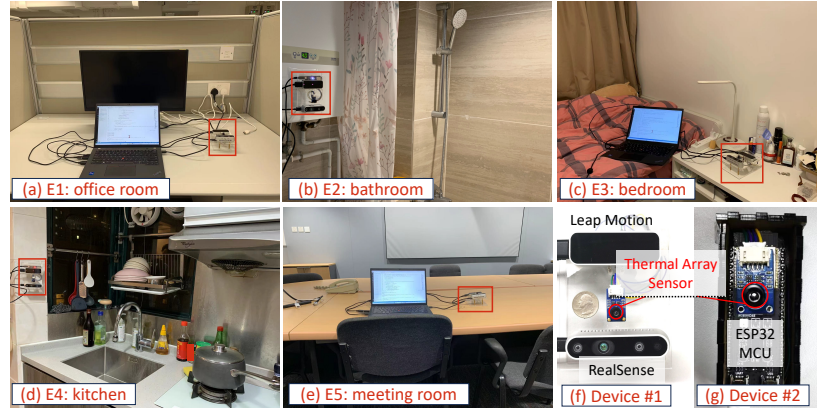


Fig. 10. Experimental environments and devices. We conduct experiments in various settings (a)-(e). (f) Device setup for data collection. (g) IoT prototype on ESP32-S3.

performance. Finally, the training loss function for NanoTapor is defined as:

$$\mathcal{L}_{NanoTapor} = \mathcal{L}_{mse}(\hat{J}, J) + \mathcal{L}_{bl}(\hat{J}, J) + \delta \mathcal{L}_{mse}(f_n, f_{super}), \quad (6)$$

where $f_{super} = A(f_{Tapor})$ is the supervision feature generated from the Tapor model, and δ is set to 0.1 empirically.

4 Implementation

Hardware: We fully implement TAPOR using the Melexis MLX90640BAB thermal array sensor, featuring a 32×24 element array with a $35^\circ \times 55^\circ$ FOV. The FOV can be expanded to $75^\circ \times 110^\circ$ with the MLX90640BAA model. The sensor's refresh rate, configurable between 0.5 Hz and 64 Hz, is set to 10 Hz for our experiments. NanoTapor runs on an ESP32-S3 board with an ESP32S3-WROOM-1 MCU.

Software: We develop the Tapor and NanoTapor models using PyTorch [49]. The NanoTapor model is then converted to TensorFlow Lite Micro [12] using the ONNX format [2] for deployment on IoT devices. We have implemented an end-to-end system for both Tapor and NanoTapor to run on diverse devices like Jetson, Raspberry PI, and ESP32.

Training: Tapor and NanoTapor models are trained using the Adam optimizer with a learning rate of 0.0001. Tapor is trained with a batch size of 48 for 200 epochs, while NanoTapor uses a batch size of 128 for 1000 epochs. All firmware, code, data, models, and demo videos are available here: <https://github.com/aiot-lab/TAPOR>.

5 Experiments

5.1 Experimental Setup

Dataset: To evaluate TAPOR, we build a dataset comprising over 60,000 arbitrary hand pose samples (*i.e.*, 32×24 temperature maps). As shown in Fig. 10, the dataset is gathered in five different environments covering different use cases: an office, a bathroom, a bedroom, a kitchen, and a meeting room. Ten participants (8 males, 2 females), aged 21–27, with hand widths from 9.4 to 13.1 cm and lengths from 17.3 to 20.5 cm, contributed to the dataset. All procedures were approved by our institution's IRB. During each trial, participants freely moved their hands above the upward-facing sensor, rotating and translating them within the FoV to cover diverse angles and positions, typically with palms facing downward in a natural interaction posture. This ensures variation in hand orientation and position while remaining within the effective sensing range. For ground truth data, we use the Leap Motion

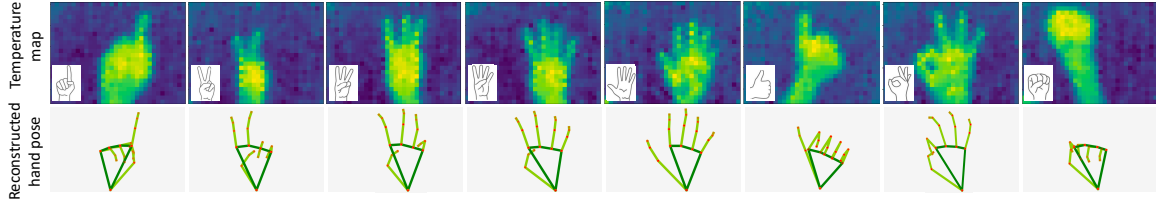


Fig. 11. Reconstructed 3D hand poses by TAPOR. The top row displays the input temperature maps alongside their corresponding gesture icons. The bottom row presents the top-down view of the reconstructed 3D hand poses.

to capture 3D hand poses, while the RealSense D435 depth camera is utilized to record depth and RGB images for our investigative experiments in §2.2. To train the Tapor and NanoTapor models, we use a subset of data from six users in the office for training, while the rest is reserved for testing. Specifically, the dataset is split into 26,557 training samples, 3,319 validation samples, and 34,520 testing samples.

Metrics: To assess the performance of TAPOR in 3D hand pose reconstruction, we employ the following three metrics:

(i) *Mean Per Joint Position Error (MPJPE)* [8, 26]: MPJPE is the average Euclidean distance between estimated key points $\hat{\mathbf{j}}$ and ground truth $\mathbf{j}$: $\text{MPJPE} = \frac{1}{J} \sum_{i=0}^J (\|\hat{\mathbf{j}}_i - \mathbf{j}_i\|_2)$, where $J = 21$ is the number of key points.

(ii) *Percentage of Correct Key points (PCK)* [79]: PCK calculates the percentage of correctly estimated joints, with a joint considered correct if its distance from the ground truth is below a specified threshold a : $\text{PCK}@a = \frac{1}{J} \sum_{i=1}^J 1(\|\hat{\mathbf{j}}_i - \mathbf{j}_i\|_2 \leq a)$, where $1(\cdot)$ is the indicator function.

(iii) *Mean Per Joint Relative Position Error (MPJRPE)*: For many tasks, including gesture recognition, the relative structure of the hand pose is adequate. Therefore, we further propose MPJRPE to measure the average Euclidean distance between estimated and ground truth joint coordinates after aligning their root points (0-th joint): $\text{MPJRPE} = \frac{1}{J} \sum_{i=0}^J (\|\hat{\mathbf{j}}'_i - \mathbf{j}'_i\|_2)$ where $\hat{\mathbf{j}}'_i = \hat{\mathbf{j}}_i - \hat{\mathbf{j}}_0$ and $\mathbf{j}'_i = \mathbf{j}_i - \mathbf{j}_0$ represent the shifted positions.

Note that lower MPJPE/MPJRPE values indicate better performance, while higher PCK signifies improved accuracy.

Baselines: TAPOR is the first 3D hand pose reconstruction system using a single thermal array sensor. To showcase the advantages of TAPOR, we compare it with three mainstream deep learning baselines in our experiments:

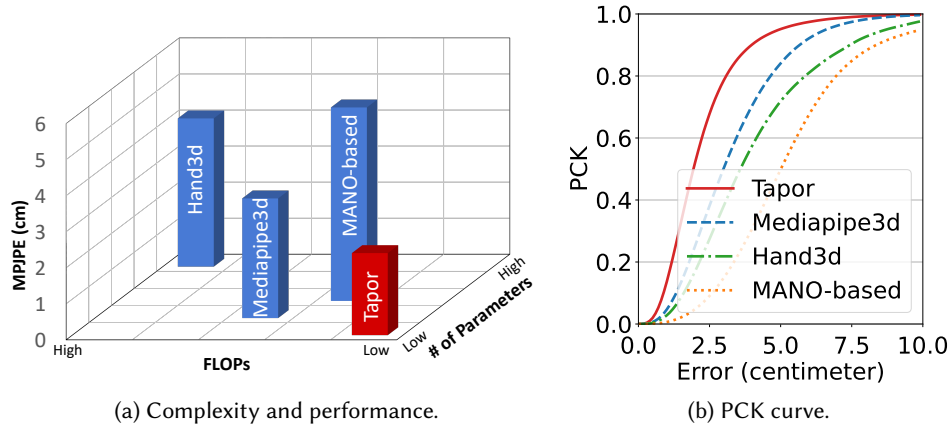
- 1) *Mediapipe3d* [83] is a well-known model that directly learns the mapping from RGB images to 3D hand poses.
- 2) *Hand3d* [92] employs a different approach by first detecting hand key points in the 2D image and then transforming these 2D key points to 3D physical space positions.
- 3) *MANO-based Model* [58] learns to map the input image to a parametric hand model derived from high-resolution 3D hand scans. The predicted parameters are then transformed into 3D key point positions using predefined rules.

All baseline models have been adapted to accept temperature maps as input. We adjust the input channel from 3 to 1 to match the single-channel temperature map and interpolate the map to match the baselines' input dimensions.

5.2 Overall Performance

We first illustrate examples of the reconstructed hand poses, as shown in Fig. 11. As seen, TAPOR reconstructs 3D hand poses accurately across a variety of gestures.

Tapor vs. Baselines: As shown in Fig. 12, Tapor outperforms the baseline models in all aspects, demonstrating the superiority of our physics-inspired design. Detailed in Fig. 12c, Tapor achieves an MPJPE of 2.26 cm and a PCK@4 cm of 90.41%. NanoTapor, though slightly less accurate than Tapor, still exceeds all the baseline models,



Method	MPJPE (cm)↓	PCK@4cm↑	FLOPs (M)↓	# of Params (M)↓
MANO-based	5.3681	0.3078	277.6867	2.2886
Hand3d	4.1137	0.5750	30103.7655	18.9612
Mediapipe3d	3.3196	0.7048	385.8148	2.0134
NanoTapor	<u>3.2683</u>	<u>0.7238</u>	0.6479	0.1565
Tapor	2.2636	0.9041	<u>244.4295</u>	<u>0.6910</u>

(c) Performance comparison (**Best**, Second best).

Fig. 12. Model comparison. NanoTapor, with its significantly lower FLOPs, is excluded in (a) for better visualization.

highlighting the effectiveness of the heterogeneous knowledge distillation technique. In terms of computational complexity, both Tapor and NanoTapor are remarkably more efficient than the baselines, with NanoTapor requiring only 0.65 million FLOPs and 0.16 million parameters.

Tapor vs. NanoTapor: NanoTapor significantly reduces the computational cost, achieving 377.3× fewer FLOPs and 4.4× fewer parameters than Tapor. This efficiency translates to minimized processing latency, as shown in Fig. 13a. On a GPU (A100), Tapor and NanoTapor report 6.63 ms and 0.28 ms latencies, respectively. Tapor runs more slowly on embedded devices Jetson Nano and Raspberry Pi 4, with a respective latency of 54.41 ms and 319.06 ms. Notably, NanoTapor significantly reduces the latencies by 16× and 85× to 3.64 ms and 3.74 ms latencies respectively, making it the only model deployable on resource-constrained IoT. Specifically, NanoTapor achieves a latency of 313 ms on ESP32-S3, which can support real-time IoT applications. Despite its lightweight design, NanoTapor’s performance in hand pose reconstruction is only slightly lower than Tapor’s but still sufficient for downstream tasks. In gesture recognition, as shown in Fig. 13b, Tapor and NanoTapor achieve F1 scores of 94.9% and 87.2%, respectively, for six gestures detailed in §5.5, indicating effective recognition capabilities. Notably, both models achieve an F1 score of 100% in the absence of a hand (Gesture #0), underscoring TAPOR’s excellent performance in avoiding false alarms, which is crucial for practical gesture control applications.

System complexity: NanoTapor requires only 1.93M of RAM and 0.502M of Flash for operation, well within the capacity of most IoT devices, such as the ESP32S3-devkitc-1 board which boasts 3.28M of RAM and 3.34M of Flash. As demonstrated in Tab. 2, activating the MLX90640 thermal array sensor and enabling the I²C interface increases power consumption by 84 mW. Additionally, running the NanoTapor system adds a further 124 mW, resulting in a

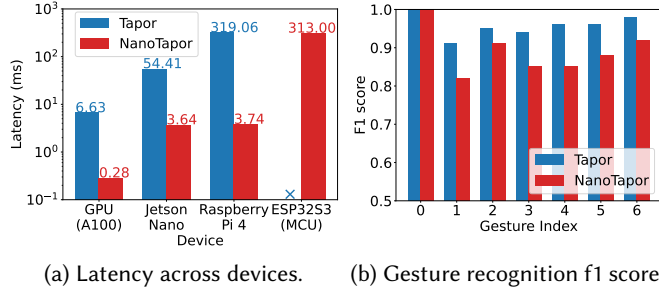


Fig. 13. Tapor vs. NanoTapor.

ESP32S3 w/ UART	MLX90640 w/ I2C	NanoTapor Operating	Power (mW)
✓	✗	✗	559
✓	✓	✗	643
✓	✓	✓	767

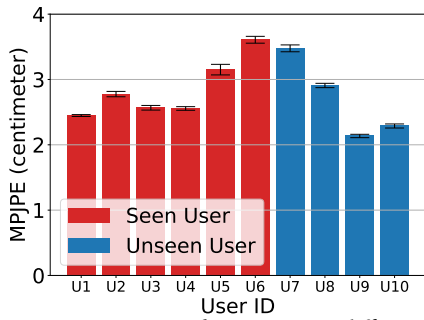
Table 2. Power consumption. The thermal array sensor with I^2C increases power by 84 mW, and the NanoTapor adds an additional 124 mW.

Fig. 14. Tapor performance on different users.

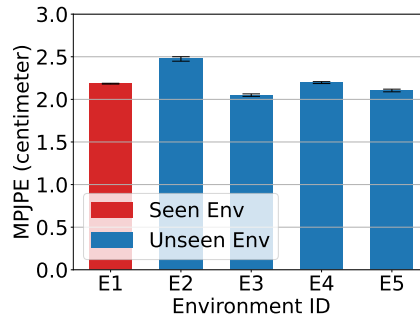


Fig. 15. Impact of different environments.

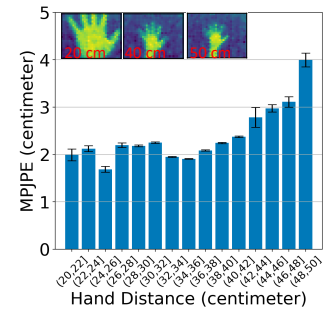


Fig. 16. Impact of hand distances.

total power consumption of 767 mW. This is significantly lower than other 3D hand pose reconstruction systems, such as the Leap Motion, which consumes over 1495 mW. It is worth noting that the majority of the power consumption is attributed to the ESP32S3 microcontroller unit (MCU) board. By transitioning to a lower-power MCU, the overall power consumption of the TAPOR system could be further reduced.

5.3 Robustness Evaluation

Cross user: For a practical 3D hand pose reconstruction system, robust performance across users is crucial. Hence, we assess Tapor's performance using data from ten participants, labeled U1-U10, by training on a subset of the data from U1-U6 and testing on the rest. As illustrated in Fig. 14, the performance discrepancies among users are within 1.5 cm MPJPE, and there is no bias towards the users seen during training, demonstrating Tapor's strong generalization ability across different users.

Cross environment: TAPOR demonstrates robust performance across various environments, as shown in Fig. 15. The Tapor model is trained with data from an office environment (E1 in Fig. 10) and then tested in all five environments. TAPOR maintains consistent performance, with MPJPE variations of less than 0.5 cm across all tested environments.

Impact of distance: We also evaluate the impact of the distance between the user's hand and the sensor, *i.e.*, depth. As depicted in Fig. 16, TAPOR maintains stable performance with MPJPE of approximately 2 cm when the distance is within 40 cm. The system performance slightly degrades, with an increase of at most 2 cm in MPJPE, when the distance ranges from 40 cm to 50 cm. This is attributed to the limited resolution of the thermal map, which hinders the separation and reconstruction of individual joints for distant hands. Despite this, the sensing range remains comparable to that of commercial hand pose estimation devices like Leap Motion, which operates

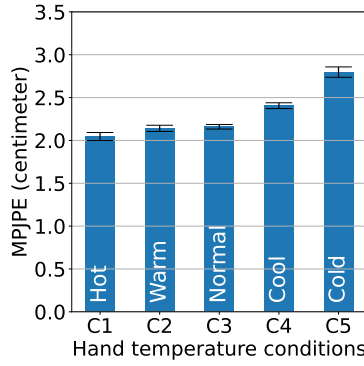


Fig. 17. Impact of hand temperature.

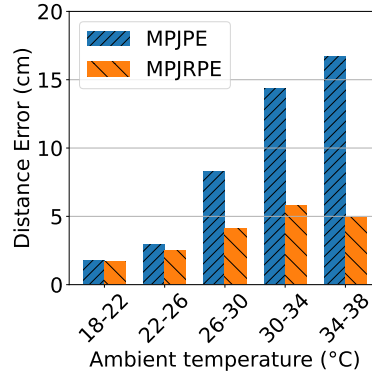


Fig. 18. Performance w.r.t. ambient temperature.

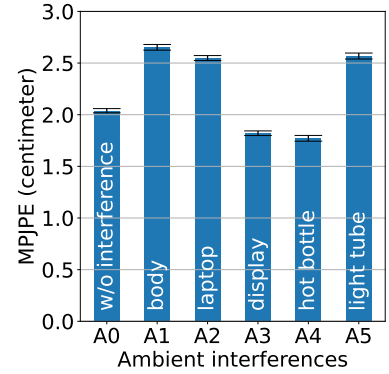


Fig. 19. Impact of ambient interference.

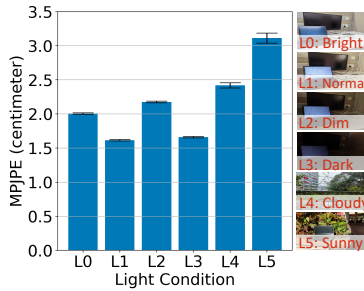
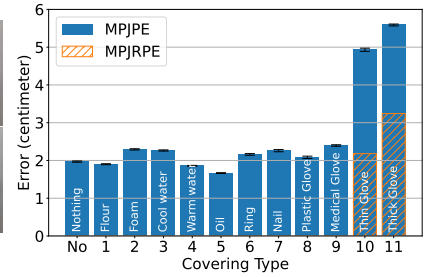


Fig. 20. Performance across different light conditions.



(a) Different hand coverings.



(b) TAPOR performances.

Fig. 21. Impact of the hand covering.

up to 60 cm using stereo vision. Overall, the MPJPE remains below 3 cm for distances up to 48 cm, sufficient for hand pose-based interaction applications primarily within the near field.

Impact of hand temperature: While human body temperatures are relatively stable, our hand surface temperatures vary depending on the environment. To evaluate the impact of hand temperature, participants immersed their hands in hot, warm, cold, and iced water for over 20 seconds before each respective experiment. The results, as Fig. 17, show robust performance with slightly increased MPJPE when the hand temperature gets lower.

Impact of ambient temperature: We evaluate performance under various ambient temperatures to study the effect of environmental thermal radiation. To isolate this factor, participants remained in rooms with different ambient temperatures for over five minutes prior to data collection, allowing the thermal background to stabilize. As shown in Fig. 18, MPJPE increases significantly beyond 26°C, while MPJRPE remains largely unaffected. This suggests that ambient temperature impacts depth estimation more than relative pose reconstruction, similar to the effect of thick gloves on thermal radiation. Despite this, with the sufficient accuracy of relative hand pose reconstruction, TAPOR can still support important applications like gesture control.

Impact of ambient interference: Certain everyday objects, such as electronic devices or heated items, may exhibit higher temperatures than their surroundings, potentially introducing interference to TAPOR. However, since TAPOR is designed for around-device interaction, the user's hand typically dominates the sensor's FoV during operation, effectively minimizing the influence of surrounding objects. To systematically evaluate TAPOR's robustness against ambient interference, we conducted experiments involving various common heat-emitting

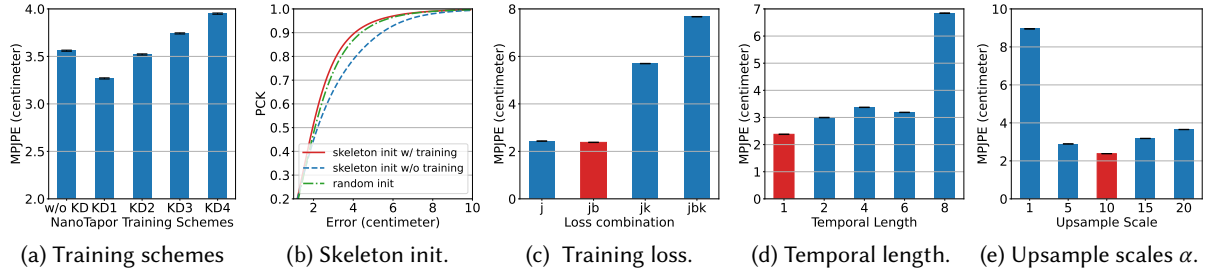


Fig. 22. Ablation study results. The ablation study is conducted on the Tapor model except for (a) on NanoTapor.

objects. These included a running laptop, an active display, a hot water bottle placed near the device, as well as the presence of the user's body and light tubes within the sensor's FoV. As illustrated in Fig. 19, the results demonstrate that TAPOR remains robust and maintains reliable performance even in the presence of such environmental interferences.

Impact of light condition: We evaluate the effect of lighting conditions on TAPOR's performance, as illustrated in Fig. 20. The results demonstrate consistent performance and robustness, even in darkness or under sunlight, with negligible variations across different light conditions.

Impact of hand coverings: As mentioned in §2, thermal array readings depend on thermal radiation emitted by the target. Hand coverings, particularly thick gloves, can partially block the radiation and affect the performance of TAPOR. Therefore, assessing the impact of different hand coverings is essential to gauge Tapor's real-world effectiveness. As in Fig. 21, Tapor performs well when the hand is covered with liquids (covering types 1 – 5), rings, and nail decorations, as these coverings do not significantly alter the thermal radiation of the hand. With gloves made from materials like plastic or latex, Tapor remains effective, as these materials allow thermal radiation to pass through. However, thick cotton gloves, which block most thermal radiation, lead to a notable increase in MPJPE. Nevertheless, the MPJRPE remains low, suggesting that the performance decline is mainly due to an underestimation of the hand's depth. Therefore, TAPOR can still accurately reconstruct the relative hand pose for various applications, including gesture recognition.

Left vs. Right Hands: We evaluate TAPOR for left and right hands separately. The MPJPE for left-hand pose reconstruction is 2.40 cm, and for right-hand pose reconstruction, it is 2.22 cm, showing no bias towards either hand.

5.4 Ablation Study

NanoTapor Training Schemes: We evaluate the performance of our heterogeneous knowledge distillation (HKD) technique compared to other methods. As shown in Fig. 22a, KD1 and KD2 use HKD with pose feature f_{pose} and location feature f_{loc} , respectively. KD3 and KD4, on the other hand, use PCA for dimension reduction with f_{pose} and f_{loc} respectively. NanoTapor performs best with HKD, outperforming PCA-based distillation and direct training.

Skeleton Init: As illustrated in Fig. 22b, Tapor with skeleton init shows the best performance, which verifies the effectiveness of injecting hand structure knowledge for training.

Loss Combination: We use joint loss (j) and bone loss (b) for Tapor optimization, as shown in Fig. 22c. Kinematic loss (k) transforms key points' positions into kinematic representation [16] and calculates MSE with ground truth. Due to poor performance, kinematic loss is not used in TAPOR.

Temporal Length: Temporal length l indicates the number of KP features used in KP fusion, where $l = 1$ uses only the current feature. As in Fig. 22d, previous features do not improve performance due to pose changes and

Ground Truth	0	782	1	1	1	1	0	0
	1	0	708	49	26	9	11	1
	2	0	18	773	5	0	1	0
	3	0	13	0	742	34	1	0
	4	0	1	0	13	732	1	0
	5	0	6	0	3	6	750	30
	6	0	0	0	0	3	0	750
		0	1	2	3	4	5	6
Prediction								

(a) Tapor.

Ground Truth	0	794	1	0	2	0	0	0
	1	0	610	60	27	17	18	5
	2	0	43	760	9	8	4	2
	3	0	24	10	654	64	5	1
	4	0	18	8	87	643	3	4
	5	0	46	10	0	16	698	35
	6	0	7	3	1	6	59	710
		0	1	2	3	4	5	6
Prediction								

(b) NanoTapor.

Fig. 23. Gesture recognition performance. We consider six gestures: 1:thumbs up, 2:raised fist, 3:crossed fingers, 4:victory hand, 5:OK hand, 6:raised hand, plus 0: No gesture.

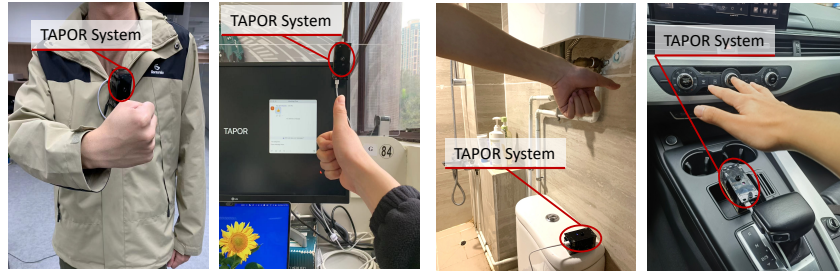


Fig. 24. Gesture control applications with TAPOR.

thermal sensor noise. We use only the current feature for optimal performance and efficiency in our prototype system.

Upsample Scale: Tapor upsamples the temperature map with a scale $\alpha = 10$, which is shown to be optimal in Fig. 22e.

5.5 Case Studies

We develop two applications, gesture control, and finger tracking, on top of TAPOR to showcase its applications.

Gesture control: To develop a gesture control system based on TAPOR, we trained a linear layer to map the reconstructed hand pose from Tapor and NanoTapor to six gestures. As in Fig. 23, both Tapor and NanoTapor achieve high accuracy for all gestures and nearly zero false alarms (Gesture #0).

With the promising performance, we envision TAPOR can enable applications in various scenarios. To demonstrate this, we conduct four case studies as shown in Fig. 24. In each case, we collect 300 samples for each of the six gestures (as in Fig. 23) and evaluate the recognition performance.

1) *Wearable AI pin control:* Recently, AI pins like Humane and Rabbit R1¹ have attracted attention for integrating AI into daily life. Though equipped with a camera and microphone, these units remain mostly inactive to save

¹Humane AI Pin: <https://humane.com/>; Rabbit R1: <https://www.rabbit.tech/>.

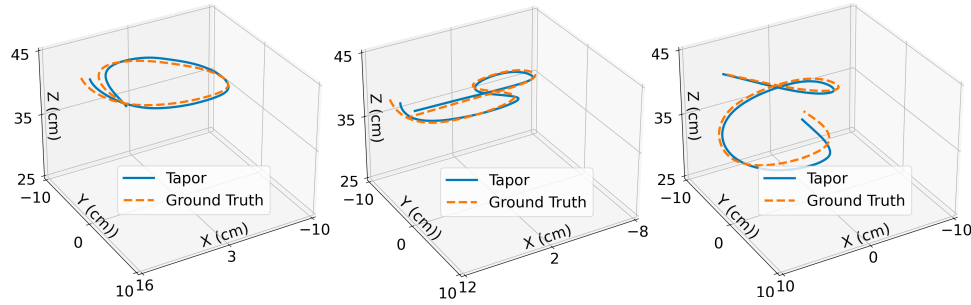


Fig. 25. Finger tracking with TAPOR.

power and protect privacy, requiring a button press for activation. TAPOR, with its lightweight design, can be integrated into wearable AI assistants to enable privacy-conscious, touch-free control with more command options than a single button.

2) *Conference calls*: Similarly, TAPOR functions as a gesture-based controller for devices like TVs and AC units. It can integrate with computers to facilitate gesture interactions in online meetings like Zoom. Users can respond with emojis, such as thumbs-up, through gestures, eliminating the need to activate the microphone or camera, or use the mouse.

3) *Toilet device control*: We also deploy TAPOR in bathrooms for operating appliances. In bathrooms, users' hands are often wet, making it risky to press electrical buttons.

4) *In-Car gesture control*: Similarly, we also showcase how TAPOR can be deployed for in-car gesture control.

Overall, Tapor achieves 97% recognition accuracy in both bathroom and car applications and maintains over 90% accuracy in the first two scenarios of wearable AI pin control and online meeting integration, while NanoTapor yields 92% and 83%, respectively. In practice, many applications often do not need six gestures (e.g., only audio on/off control for AI pins). Therefore, we further evaluate the accuracy by varying the gesture vocabulary. Our results show that NanoTapor achieves an accuracy of 93% for 4 gestures and 98% for 2 gestures, with random combinations from the six gestures.

Finger Tracking: Finger tracking is a highly sought-after capability with numerous practical applications, garnering significant attention in both research and industry. Leveraging its robust 3D hand pose reconstruction functionality, TAPOR demonstrates exceptional precision in tracking intricate finger movements within three-dimensional physical space. As depicted in Fig. 25, TAPOR effectively tracks the random motion of an index finger, achieving an average position error of just 2.69 cm across 40 trials when combined with a Kalman filter for post-processing. This outstanding performance underscores TAPOR's potential as a powerful tool for advanced finger-tracking applications.

6 Discussions and Future Works

System cost: The TAPOR prototype costs around US\$20 for a thermal sensor and US\$5 for the ESP32-S3 DevKit. The cost can be further cut down when building in bulk.

Thermal depth: TAPOR's performance, especially depth accuracy, can be impacted by ambient temperatures and target radiation, a well-known issue in thermal sensing [13, 68]. In its current form, TAPOR maintains high precision at typical indoor temperatures (18°C-26°C) and low pose error at higher temperatures. Future work will explore techniques like background thermal cancellation for improvement.

Better accuracy: It remains an open question whether thermal array-based methods can achieve hand pose estimation accuracy comparable to RGB camera-based systems. Exploring approaches such as stereo thermal arrays or multi-sensor fusion may help close this gap and is an important direction for future work.

Hand orientation failure: Cases in which the palm faces upward (*i.e.*, away from the sensor) are rare but represent a potential failure mode due to limited training data for such scenarios. Since our work focuses on around-device interactions, we primarily consider common postures where the palm naturally faces downward toward the sensor, as is typical in interaction scenarios.

Multi-hand pose: Currently, TAPOR supports the reconstruction of single-hand poses, which is sufficient for the majority of interaction scenarios. Notably, many existing gesture control systems, such as those in the Apple Watch, Google Soli [37], and Vtouch [74] commercial systems, are also designed for single-hand use. However, extending TAPOR to support multi-hand and multi-user interactions presents an exciting avenue for future research, enabling more complex and collaborative applications.

Close-range interactions: TAPOR is designed for close-range interactions, similar to systems like Leap Motion [71] and Google Soli [37]. TAPOR's around-device interaction capabilities hold significant potential for a wide range of applications. For instance, users can employ customized or preferred gesture sets to control TVs, in-car media systems in low-light conditions, or navigate maps on smartwatches. Additionally, TAPOR can be used for AR/VR headset interaction, controlling smart appliances in the kitchen with a wet hand, and gesture control in the absolute dark. Mid- and long-range interactions remain an area for future exploration.

Hand-Object interactions: This work focuses on free-hand pose estimation for around-device interaction. However, hand pose estimation while holding objects is useful for applications in virtual reality (VR) and augmented reality (AR). Investigating this capability will be a key focus of our future research, as it opens up new possibilities for immersive and interactive experiences.

7 Related Works

7.1 3D Hand Pose Estimation

3D hand pose reconstruction has been widely studied. We mainly survey the literature using the below modalities.

Camera-based: These methods predominantly utilize RGB(D) images for hand pose estimation and shape reconstruction. Our focus is on monocular 3D hand estimation methods, which can be categorized into three categories: 1) Direct regression methods estimate 3D hand joint positions from RGB(D) images using end-to-end neural networks [33, 48, 83]. However, these methods face challenges due to limited labeled 3D hand datasets and insufficient supervision, leading to research on lifting 2D positions to 3D. 2) 2D lifting approaches detect key points in 2D frames and then convert these to 3D positions [54, 67]. 3) Model-based methods leverage prior knowledge of hand structure to enhance estimation accuracy [8, 61, 91], with MANO model [58] being a foundational model in this category with many followers like HTML [53], NIMBLE [36], DART [18]. Despite their accuracy, camera-based approaches face challenges like privacy concerns and sensitivity to lighting conditions.

Signal-based: Wireless, acoustic, and light signals are exploited as more privacy-friendly alternatives for hand pose reconstruction, mainly focusing on gesture recognition [9, 31, 50, 52, 62] or hand/finger tracking [32, 37, 41, 44, 63, 75]. Among these, Aili [34] reconstructs hand skeletons by capturing the blockage of light from a custom LED-panel-and-photodiode-array setup, while HandFi [27] leverages WiFi signals to infer 3D hand poses using the existing communication infrastructure. Despite their creativity, these systems face challenges: Aili relies on bulky, non-integrable hardware not suited for compact IoT environments, while HandFi suffers from weak boundary localization and generalization limitations, which may cause unintended activations in real-world settings.

Wearable-based: Involving various sensors attached to the wrist, fingers, or integrated into gloves, these methods are used for gesture recognition [28, 29, 40, 78] and hand pose estimation [76, 90]. For instance, MagX

[7] enables untethered hand tracking by detecting magnetic fields from passive magnet rings worn on fingers, while RoFin [84] employs custom gloves with embedded LED nodes to achieve fine-grained, real-time 3D pose reconstruction. In a similar vein, Pyro [20] introduces a compact, low-power thumb-tip gesture recognition system using pyroelectric sensors. Its design demonstrates robustness across lighting conditions and motion scenarios, and its small form factor makes it attractive for wearable and mobile use.

However, despite their performance, all these wearable-based systems require users to wear specific devices, which can be intrusive or uncomfortable over extended periods. This makes them less suitable for ambient or large-scale IoT deployments, where ease of use, unobtrusiveness, and seamless integration are critical.

7.2 Thermal Array-based Sensing

Thermal array sensors are gaining prominence due to their versatility in a variety of sensing applications. These applications include human detection [6, 65], occupancy estimation [10, 59, 70], daily activity recognition [43, 51, 57], fall detection [21, 46, 55, 88], and indoor localization [4, 17, 22]. For gesture recognition, although there has been extensive work using thermal cameras, such as Birkeland *et al.*[3], who provided a large thermal hand gesture dataset captured with the FLIR Lepton thermal camera, and ThermoHands [15], which introduced an egocentric benchmark dataset and a transformer-based baseline for robust 3D hand pose estimation under diverse real-world conditions, research on thermal array sensors remains limited.

As a new and emerging sensing modality, thermal arrays offer unique advantages such as extremely low resolution, low signal-to-noise ratio, low power consumption, and significantly lower cost compared to thermal cameras. Despite these promising features, only a few studies have explored their potential for gesture recognition or hand pose estimation. Notable efforts in this direction include Tatenno *et al.*[66], who developed an in-car interface system using a thermal array sensor and a CNN model, and Vandersteegen *et al.*[72], who proposed a low-latency gesture recognition system based on temporal convolutional networks applied to thermal array map sequences. A particularly relevant study, FingerTrak [24], employs multiple thermal array sensors embedded in a wristband to continuously track hand poses. While this design effectively addresses occlusion issues, it introduces significant hardware complexity and imposes wearability constraints. In contrast, TAPOR is tailored for ambient, non-wearable scenarios. It leverages a single low-cost thermal array sensor to enable seamless around-device interaction, significantly simplifying deployment and enhancing usability.

TAPOR distinguishes itself by enabling 3D hand pose reconstruction solely from a thermal array sensor, offering a scalable, privacy-preserving, and cost-effective solution well-suited for widespread IoT applications.

8 Conclusions

We present TAPOR, an accurate and efficient 3D hand pose reconstruction system for around-device interaction applications using a single low-cost thermal array sensor. TAPOR achieves this by contributing a physics-inspired neural network that learns from thermo-depth physics, 2D thermo-pose context, and hand structure knowledge. TAPOR also presents a novel heterogeneous knowledge distillation technique that enables NanoTapor, an ultra-lightweight model suitable for real-time IoT adoption. We conduct extensive real-world experiments to demonstrate TAPOR's high accuracy with low complexity, and carry out case studies on gesture control and finger tracking to showcase its potential applications. TAPOR sheds light on a ubiquitous around-device interaction interface for the Internet of Thermal Things.

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